

Selfish, Local and Online Scheduling via Vector Fitting

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Introduction

A technique, applicable to scheduling/congestion problems with a quadratic objective function C , allowing to upper bound

$$\frac{C(x)}{\text{OPT}}$$

where x can be a

- Nash equilibrium ($\rightarrow$ *price of anarchy*)
- Local optimum ($\rightarrow$ *approximation ratio of local search algorithms*)
- Output of an online algorithm ($\rightarrow$ *competitive ratio*)

Example: Load balancing

Input: Set of machines M and jobs N with weights $w_{ij} \in \mathbb{R}_+ \cup \{\infty\}$.

Goal: Find an assignment $x \in \{0, 1\}^{M \times N}$ of jobs to machines to minimize

$$C(x) = \sum_{i \in M} \ell_i(x)^2$$

where $\ell_i(x) = \sum_{j \in N} w_{ij} x_{ij}$ is the *load* of a machine.

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Game-theoretic setting: each job selfishly picks one of its feasible machines $\mathcal{S}_j \subseteq M$ and wants to minimize its own share

$$C_j(x) = \sum_{i \in M} \ell_i(x) w_{ij} x_{ij}$$

Online setting: each job arrives one by one and needs to irrevocably be assigned to a machine by an algorithm

Example: $R|| \sum w_j C_j$

Input: M , N , processing times $p_{ij} \in \mathbb{R}_+ \cup \{\infty\}$, weights $w_j \in \mathbb{R}_+$

Goal: Find assignment x (and ordering on each machine) to minimize

$$C(x) = \sum_{j \in N} w_j C_j(x)$$

where $C_j(x)$ is the completion time of $j \in N$.

Game-theoretic setting: each job needs to pick one of its feasible machines $\mathcal{S}_j \subseteq M$ in order to minimize its own completion time $C_j(x)$.

Online setting: each job arrives one by one and needs to irrevocably be assigned to a machine by an algorithm

Price of Anarchy

Definition

An assignment x is a *pure Nash equilibrium* if

$$C_j(x) \leq C_j(x_{-j}, i) \quad \forall j \in N, \forall i \in \mathcal{S}_j.$$

Price of Anarchy (PoA)

The *price of anarchy* of a game is the worst-case, over all instances, of

$$\frac{C(x)}{\text{OPT}} \in [1, \infty]$$

where x is any Nash equilibrium.

Obtainable results

For $R|| \sum w_j C_j$, changing the ordering policy on each machine can improve the price of anarchy.

Paper: *Inner product spaces for MinSum coordination mechanisms*
[CCGMO, STOC 2011]

Price of anarchy for $R|| \sum w_j C_j$

- *Smith's Rule* leads to a PoA of 4
- A preemptive policy called *Proportional Sharing* leads to a PoA of $(3 + \sqrt{5})/2 \approx 2.618$
- A randomized policy called *Rand* leads to a PoA of 2.133

Other PoA results

- Price of anarchy of weighted affine congestion games
- Pure price of anarchy of $P|| \sum w_j C_j$

Local search for $R|| \sum w_j C_j$

- Move a job from one machine to another if that improves the objective function gives an approximation ratio of ≈ 2.618 [CM22]
- Best known combinatorial approximation algorithm based on local search achieves a ratio of ≈ 1.809 [CGV17]

Online algorithms for $R|| \sum w_j C_j$

- One deterministic (greedy) algorithm [GMUX20] and another randomized algorithm achieving the optimal competitive ratio of 4

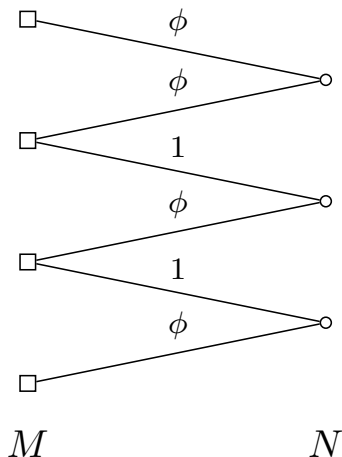
Follow-up work

Paper: *Improved online load balancing in the two-norm* [Borst, K. 25]

- Greedy algorithm is ≈ 5.828 -competitive [AAG+95], randomized algorithm based on independent rounding is 5-competitive [Car08].
- Primal-dual randomized algorithm achieving ≈ 4.98 using negative correlation rounding

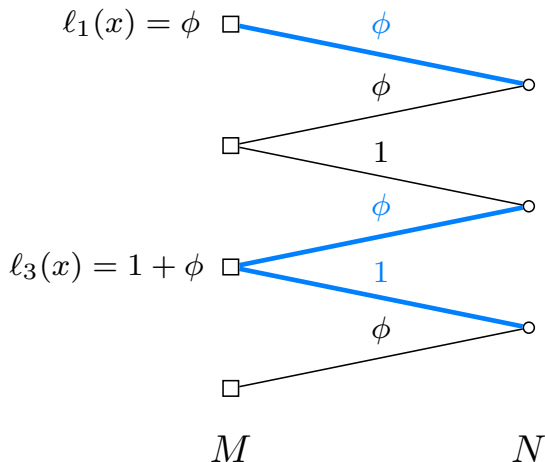
Example: load balancing

Let $\phi \approx 1.618$ such that $\phi^2 = 1 + \phi$. Instance with $|M| = |N| + 1 = n + 1$



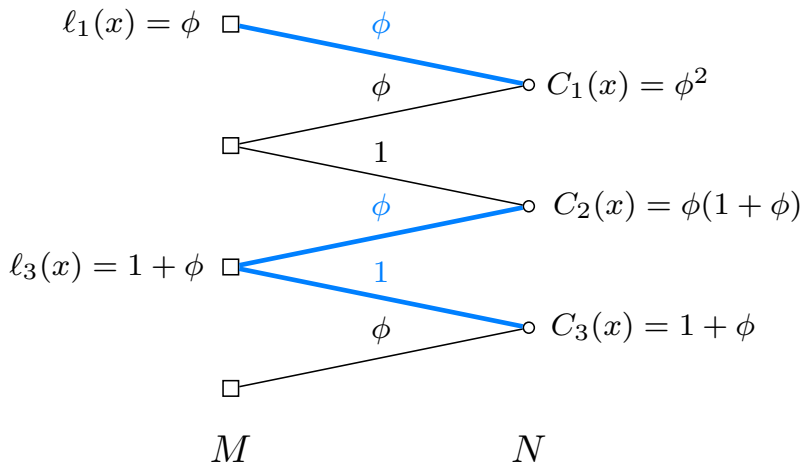
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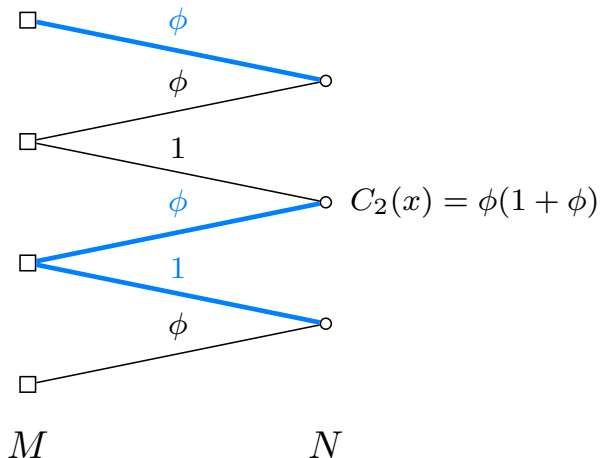
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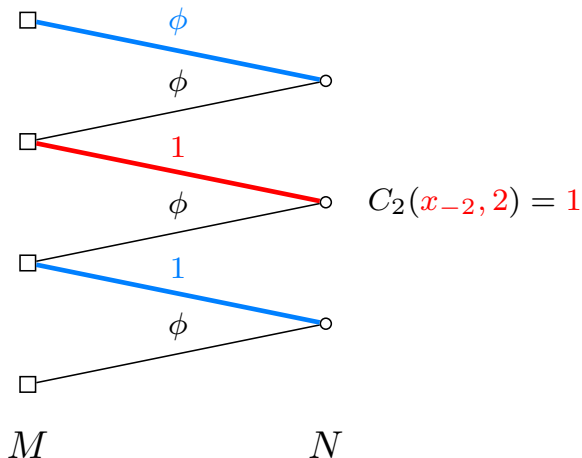
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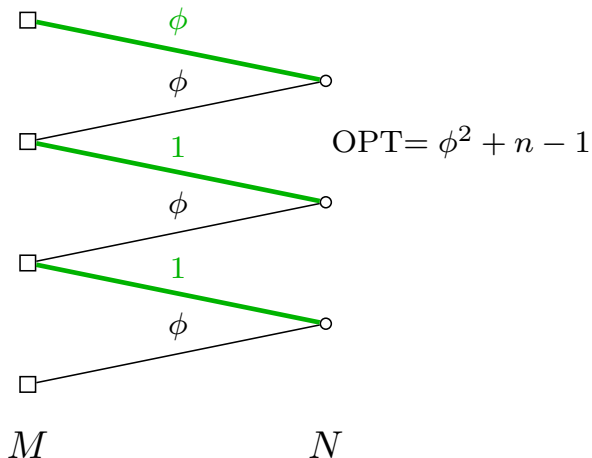
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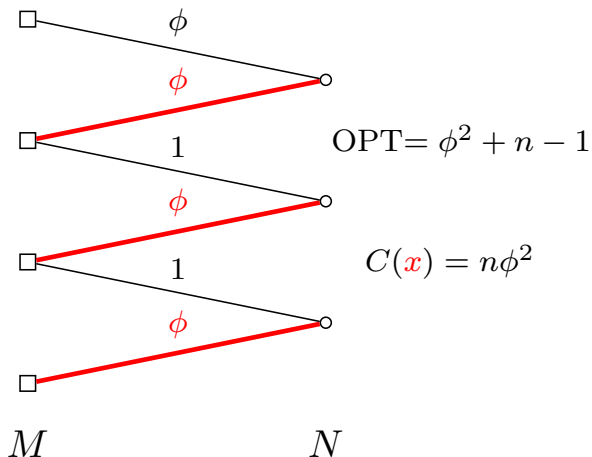
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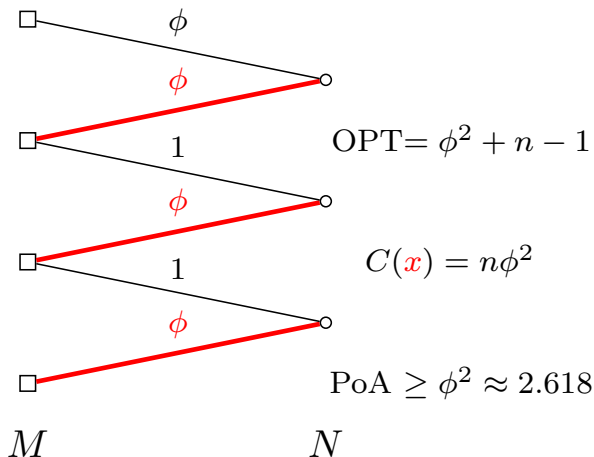
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Dual fitting: high-level view

Exact binary quadratic program to compute OPT:

$$\begin{aligned} \min C(x) \\ \sum_{i \in \mathcal{S}_j} x_{ij} &= 1 \quad \forall j \in N \\ x_{ij} &\in \{0, 1\} \quad \forall j \in N, \forall i \in \mathcal{S}_j. \end{aligned}$$

Goal: [Kulkarni-Mirrokn, 2014]

For any Nash equilibrium x , find a feasible dual solution to some convex relaxation with objective value V such that

$$V \geq \rho C(x) \quad \text{for some} \quad \rho \in [0, 1]$$

$$\implies \text{PoA} = \frac{C(x)}{C(x^*)} \leq \frac{C(x)}{V} \leq \frac{1}{\rho}$$

Convex SDP relaxation

Since $C(x)$ is quadratic, then

$$C(x) = \langle C, X \rangle := \text{Tr}(CX)$$

for some symmetric matrix C , where $X = (1, x)(1, x)^T$ encodes all the linear and quadratic terms of x .

A semidefinite programming relaxation:

$$\min \langle C, X \rangle$$

$$\sum_{i \in \mathcal{S}_j} X_{\{ij, ij\}} = 1 \quad \forall j \in N$$

$$X_{\{0,0\}} = 1$$

$$X_{\{0, ij\}} = X_{\{ij, ij\}} \quad \forall j \in N, i \in \mathcal{S}_j$$

$$X \succeq 0$$

The dual SDP

Variables:

- Scalars $y_j \in \mathbb{R}$ for every $j \in N$
- Vector $v_0 \in \mathbb{R}^M$
- Vectors $v_{ij} \in \mathbb{R}^M$ for every $j \in N, i \in \mathcal{S}_j$

$$\max \sum_{j \in N} y_j - \frac{1}{2} \|v_0\|^2$$

$$y_j \leq C_{\{ij, ij\}} - \frac{1}{2} \|v_{ij}\|^2 + \langle v_0, v_{ij} \rangle \quad \forall j \in N, i \in \mathcal{S}_j$$

$$\langle v_{ij}, v_{i'k} \rangle \leq 2 C_{\{ij, i'k\}} \quad \forall (i, j) \neq (i', k) \text{ with } j, k \in N$$

Dual fitting: high-level view

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- *Price of anarchy*: make (1) correspond to Nash equilibria inequalities

$$C_j(x) \leq C_j(x_{-j}, i) \quad \forall j \in N, i \in \mathcal{S}_j.$$

- *Local search*: make (1) correspond to local optima inequalities
- *Online algorithms*: make (1) correspond to inequalities satisfied by an online algorithm at every time step

Back to our example: load balancing

Specialize the SDP:

$$\max \sum_{j \in N} y_j - \frac{1}{2} \|v_0\|^2$$

$$y_j \leq w_{ij}^2 - \frac{1}{2} \|v_{ij}\|^2 + \langle v_0, v_{ij} \rangle \quad \forall j \in N, i \in \mathcal{S}_j$$

$$\langle v_{ij}, v_{i'k} \rangle \leq 2 w_{ij} w_{ik} \mathbb{1}_{\{i=i'\}} \quad \forall (i,j) \neq (i',k) \text{ with } j, k \in N$$

Nash equilibria inequalities:

$$C_j(x) \leq w_{ij}^2 + w_{ij} \ell_i(x) \quad \forall j \in N, \forall i \in \mathcal{S}_j.$$

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Nash equilibria inequalities:

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Fitting:

- $v_{ij}(i') = \alpha w_{ij} \mathbb{1}_{\{i=i'\}}$ for some $\alpha \in [0, \sqrt{2}]$
- $v_0(i) = \beta \ell_i(x)$ for some $\beta \geq 0$
- $y_j = \gamma C_j(x)$ for some $\gamma \geq 0$

SDP constraints:

$$\begin{aligned} y_j &\leq w_{ij}^2 - \frac{1}{2} \|v_{ij}\|^2 + \langle v_0, v_{ij} \rangle \\ &\iff \gamma C_j(x) \leq \left(1 - \frac{\alpha^2}{2}\right) w_{ij}^2 + \alpha\beta w_{ij} \ell_i(x) \end{aligned} \quad (2)$$

Constraints on the constants: $\gamma = 1 - \frac{\alpha^2}{2} = \alpha\beta$

SDP objective: $\sum_{j \in N} y_j - \frac{1}{2} \|v_0\|^2 = \left(\alpha\beta - \frac{\beta^2}{2}\right) C(x)$

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Solve:

$$\max \left\{ \alpha\beta - \frac{\beta^2}{2} : \alpha\beta = 1 - \frac{\alpha^2}{2}, \alpha \in [0, \sqrt{2}], \beta \geq 0 \right\} = \frac{2}{3 + \sqrt{5}} = \frac{1}{\phi^2}$$

Extension to congestion games

Also works for affine *congestion games*, where $S_j \subseteq 2^M$.

Conclusion

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- Unified proof technique for scheduling and congestion problems whose optimal solution can be cast as a binary quadratic program
- SDP relaxation can be obtained by the first round of Lasserre/SoS hierarchy
- Recovers and unifies numerous results
- Works in game-theoretic, local search and online settings

Future work ideas

- Apply this technique to new problems with a quadratic objective
- Design a scheduling policy for $R|| \sum w_j C_j$ improving PoA of ≈ 2.133
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Thanks!